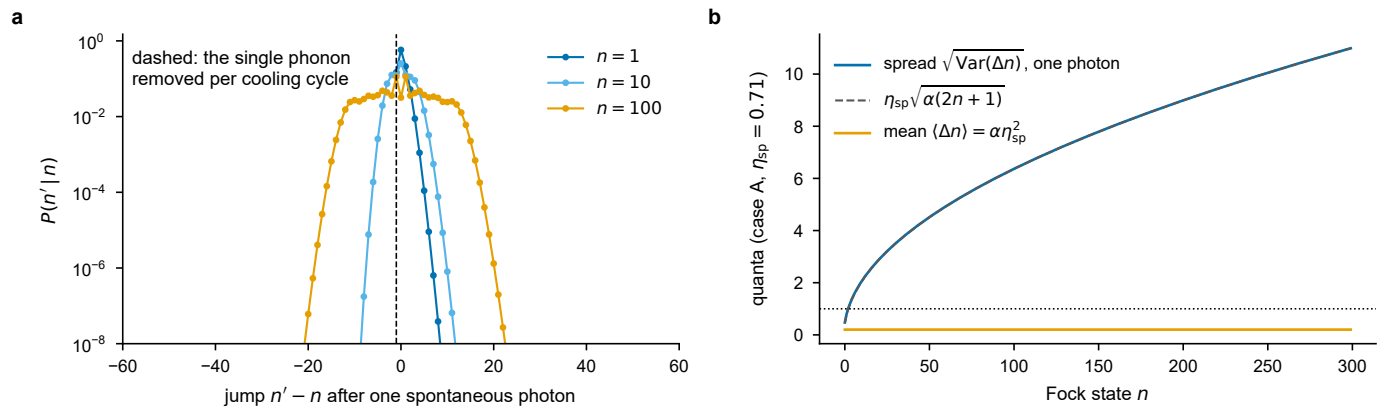

a
dashed: the single phonon
removed per cooling cycle
$n = 1$
$n = 10$
$n = 100$
$P(n' \mid n)$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
−60
−40
−20
0
20
40
60
jump $n' - n$ after one spontaneous photon
b
spread $\sqrt{\mathrm{Var}(\Delta n)}$, one photon
$\eta_{\mathrm{sp}}\sqrt{\alpha(2n+1)}$
mean $\langle \Delta n \rangle = \alpha\eta_{\mathrm{sp}}^2$
quanta (case A, $\eta_{\mathrm{sp}} = 0.71$)
0
2
4
6
8
10
0
50
100
150
200
250
300
Fock state n